

Drava Runtime System

Application Interface
Python / C API Bindings

Execution Engine

Pipeline Configuration

Stage definitions, Transport type, Ingress/egress routing, Runtime parameters

Dispatch and Batching

Frame accumulation, callback grouping

Atomic Metrics Counters

Throughput, latency, callback timing

Work-Stealing Thread Pool (XKRT)

I/O Thread

Transport fetch loop

Spawn
Tasks →

Compute Threads (1..N)

Execute callback tasks

↓ Pull messages

Transport Abstraction

Unix Socket
Local IPC

NATS Jetstream
Distributed system